Section 1 Lab Objectives

When this lab exercise is completed, the student should be able to:

1. Explain the difference between linear and switching regulators.
2. Design a linear regulator.
3. Use a switching regulator IC.

Section 2 Logistics

This lab exercise is in two parts scheduled over two weeks.

1. Complete the pre-lab before coming to lab in Section 3.
2. Follow the instructions in Section 7 on report submission, due one week after completing both parts.

Section 3 Pre-Laboratory Preparation

Prior to your scheduled laboratory meeting time the following items need to be completed.

On Line Learning

1. Watch *Buck converter vs. linear voltage regulator - practical comparison* at `https://www.youtube.com/watch?v=giGRrODKJSE`
2. Watch *EEVblog #90 - Linear and LDO regulators and Switch Mode Power Supply Tutorial* at `https://www.youtube.com/watch?v=cM7t1Mpu7s4`
3. Watch *Switching Voltage Regulator (Buck, Boost) Introduction — AO #18* at `https://www.youtube.com/watch?v=CPvA-uSOdNY`
4. Watch *Boost Converters and Buck Converters: Power Electronics* at `https://www.youtube.com/watch?v=vwJYIorz_Aw`



Voltage dividers are an important *part* of a voltage regulator. If needed, review them by watching *Make a Voltage Divider - Simply Making* at `https://www.youtube.com/watch?v=L321aqTvrLQ`. As explained in *Voltage Dividers as Regulators ?! — AO #22* at `https://www.youtube.com/watch?v=-kEh0TYjYYE`, voltage dividers are part of a regulator and cannot do the job all by themselves.

Section 4 Lab Procedure - Linear Regulator Build - Week 1

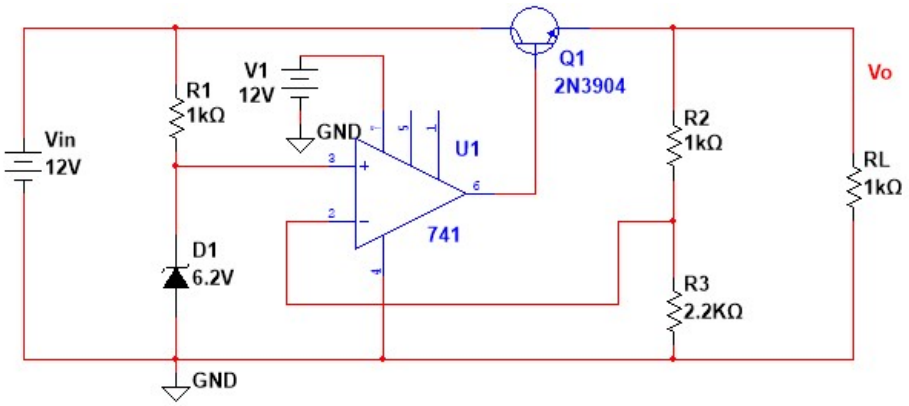


Figure 1: Operational Amplifier Linear Regulator

Figure 1 is a linear regulator built with an operational amplifier that contains the basic elements of any voltage regulator. It has a voltage divider with a zener, creating the reference voltage and an op-amp controlling a NPN transistor to regulate the output voltage.

Build and Verify

1. Build the circuit in Figure 1 using a *1N4735 zener diode* for D_1.

> One of the most important engineering skills is troubleshooting. When your circuit doesn't work, try to fix it and resist the urge to just rebuild it.

2. Verify the following:
   - 12V on pin 7 and 0V on pin 4.
   - $6.2V \pm 5\%$ on pin 3.
   - The voltage on pin 2 and 3 is equal.

> Missing $12V$ or ground to the op-amp and a bad op-amp are the most common reasons for the voltage on pins 2 and 3 being different.

   - V_O.

> V_O can be found by algebraically manipulating the voltage divider equation. V_O can also be found by using the known voltage across R_3 to determine the current in both R_2 and R_3 and Ohm's Law.

Important Characteristics

1. Measure the input current provided by V_{IN} and V_1. Calculate the input power by adding the power provided by both sources.

2. Measure the output current provided to R_L and calculate the output power.
3. Calculate the efficiency ($\frac{P_{OUT}}{P_{IN}}$) and convert it to a percentage (expect 25-50% or less).



Input power includes the power from both V_{IN} and V_1.

4. Measure the output using the oscilloscope using AC Coupling. Measure the peak-peak ripple (will look like random noise).

Section 5 Lab Procedure - Switching Regulator Test - Week 1

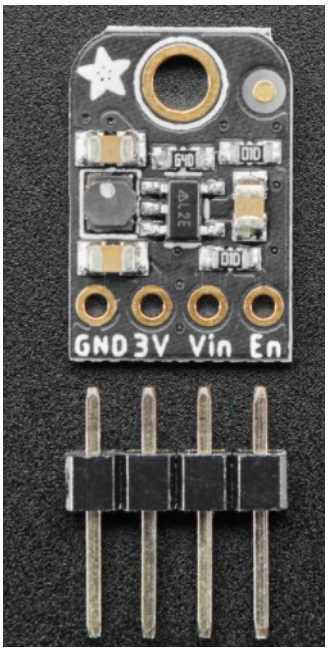


Figure 2: AP3429a Switching Regulator

Figure 2 shows a switching regulator board running at a fixed 1-MHz. The chip can provide 2A but the inductor on the board only supports 1.2A.

Power and Verify

1. Plug the AP3429a into your breadboard.
2. Provide 5V to V_{IN} and connect ground. ***NOT 12V!***
3. Connect a $1K\Omega$ load resistor (R_L) from the 3V pin to the GND pin.
4. Connect En (enable) to 5V.
5. Verify the 3V pin is around 3.3V.

Measure Important Characteristics

1. Measure the input current provided by V_{IN} and calculate the input power.
2. Measure the output current provided to R_L and calculate the output power.

3. Calculate the efficiency ($\frac{P_{OUT}}{P_{IN}}$) and convert it to a percentage (expect 85% or more).
4. Measure the output using the oscilloscope using AC Coupling. Measure the peak-peak ripple (will look like random noise).

Section 6 Lab Procedure - Switching Regulator using TI's WEBENCH - Week 2

Design and simulate a switching regulator using *TI's WEBENCH POWER DESIGNER*.

Design

1. Go to `https://webench.ti.com/power-designer/switching-regulator`.
2. Enter Input (DC) minimum 3.4V and maximum 5.5V.
3. Enter Output 3.3V at 1.2A.
4. Select view designs.
5. Select *Module (Integrated Inductor)* as the regulator type and find the LMZ10503 design.



WEBENCH can simulate the design after you register. I have had a TI account for years, TI makes lots of good ICs. We will use Multisim instead.

Verify

1. Enter the design in Multisim. Use a $1K\Omega$ load resistor (R_L) and *LMZ10503EXTTZ/NOPB* for the regulator.
2. Add voltage markers and verify the output voltage when the input is 4V and 5V. A partial output is shown in Figure 3.



The simulator will take several minutes to calculate the correct output voltage. If you look at the bottom right corner the simulation time is shown. Note that the circuit is complicated enough to take a few minutes to simulate only a few milliseconds. This is typical of real designs used in industry.

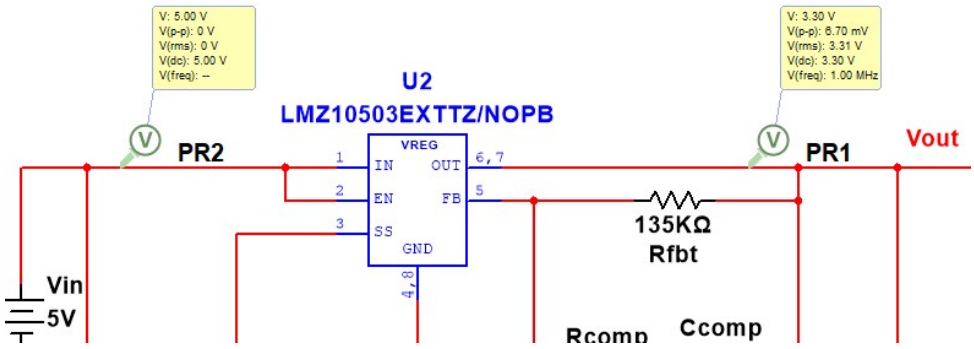


Figure 3: Partial schematic showing LMZ10503 design simulation with $V_{IN} = 5V$

Section 7 Report - Submit at start of next lab

1. Submit (any) cover page with name and major. Memes appreciated.
2. Measured Characteristics (power, efficiency and ripple) for the linear and switching regulators.
3. Simulation of the LMZ10503 regulator showing voltage markers (both 4V and 5V).